

KUNDA, Estonia

WMO: 260450

Lat: 59.5211N

Lon: 26.5417E

Elev: 4

StdP: 101.27

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.6	-16.0	-22.2	0.5	-19.2	-18.8	0.7	-15.5	14.5	-0.2	12.8	0.1	2.1	160	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	26.2	19.4	24.1	18.7	22.4	17.8	20.5	24.4	19.4	23.0	18.4	21.7	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.8	18.0	12.9	21.4	16.9	12.1	20.5	59.0	24.3	55.2	23.1	51.9	21.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.8	9.5	-21.4	29.4	4.7	2.4	-24.8	31.1	-27.5	32.5	-30.1	33.9	-33.5	35.6		
(5)			-22.1	21.9	4.3	1.4	-25.2	22.9	-27.7	23.8	-30.1	24.5	-33.3	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-3.9	-4.5	-1.2	4.4	9.6	14.1	17.5	16.4	12.2	6.7	1.9	-1.4
	DBStd	8.65	5.59	6.08	4.60	3.96	4.12	3.00	2.88	2.84	3.12	3.58	3.91	4.96
	HDD10.0	2139	432	404	347	175	58	2	0	0	14	112	243	352
	HDD18.3	4530	690	638	605	418	274	132	51	76	183	360	493	611
	CDD10.0	697	0	0	0	7	44	126	232	197	82	9	0	0
	CDD18.3	47	0	0	0	0	2	5	25	14	1	0	0	0
	CDH23.3	263	0	0	0	0	14	37	134	75	3	0	0	0
	CDH26.7	48	0	0	0	0	2	6	26	15	0	0	0	0
Wind	WSAvg	4.4	4.9	4.8	4.5	4.0	3.7	4.0	3.7	3.8	4.2	4.8	4.8	5.3
Precipitation	PrecAvg	579	40	30	28	26	43	66	62	75	55	59	56	39
	PrecMax	822	82	76	57	60	117	170	115	179	102	120	94	64
	PrecMin	395	7	3	8	4	2	14	3	9	18	14	9	10
	PrecStd	104	18	16	15	13	31	39	31	44	27	24	20	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	5.9	10.5	20.1	25.1	27.3	29.8	28.6	22.7	16.1	10.5	8.9
		MCWB	4.7	4.0	5.7	11.5	17.3	18.5	21.3	19.3	16.5	12.6	9.2	7.2
	2%	DB	4.1	4.4	7.5	15.7	21.0	23.7	26.5	25.3	20.0	14.1	9.3	7.0
		MCWB	3.2	3.2	4.1	9.5	14.3	17.6	20.2	18.6	15.6	12.1	8.2	5.8
	5%	DB	3.1	3.4	5.4	12.8	18.3	21.2	24.5	23.0	18.4	12.9	8.3	5.6
		MCWB	2.3	2.4	3.0	8.0	13.1	16.4	19.5	18.1	14.8	11.3	7.2	4.5
	10%	DB	2.2	2.4	3.9	10.3	16.0	19.3	22.9	21.3	17.0	11.7	7.1	4.3
		MCWB	1.6	1.5	2.2	6.8	11.6	15.1	18.7	17.4	14.2	10.3	6.2	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	4.4	6.4	12.2	18.2	20.1	22.3	21.2	17.7	13.5	9.5	7.6
		MCDB	5.5	5.3	9.8	17.9	23.5	24.8	27.1	25.1	20.5	14.8	10.2	8.6
	2%	WB	3.3	3.4	4.6	10.2	15.5	18.4	21.1	19.6	16.4	12.5	8.3	5.9
		MCDB	3.9	4.3	6.9	14.8	19.6	22.4	25.2	23.3	19.2	13.8	9.2	6.9
	5%	WB	2.5	2.5	3.4	8.5	13.8	17.1	20.1	18.6	15.4	11.4	7.3	4.6
		MCDB	3.1	3.4	5.0	12.1	17.2	20.5	23.8	22.3	17.8	12.8	8.1	5.5
	10%	WB	1.6	1.6	2.5	7.0	12.1	15.8	19.1	17.7	14.5	10.4	6.3	3.4
		MCDB	2.2	2.4	3.7	10.0	15.4	18.9	22.3	21.0	16.5	11.6	7.1	4.2
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	5.8	7.1	8.0	7.3	7.4	7.6	6.5	4.8	3.5	3.6
		MCDBR	3.4	4.2	7.3	11.6	11.8	11.0	10.1	10.2	8.5	6.0	4.3	3.6
		MCWBR	3.3	3.6	4.9	6.6	6.4	6.0	5.3	5.3	5.0	4.2	3.9	3.3
	5% WB	MCDBR	3.3	4.1	6.2	10.6	10.5	9.8	9.3	8.9	7.8	5.5	4.2	3.5
		MCWBR	3.2	3.7	4.5	6.4	6.3	6.0	5.3	5.4	5.2	4.2	3.9	3.3
Clear-Sky Solar Irradiance	taub	0.276	0.308	0.332	0.373	0.370	0.367	0.389	0.387	0.356	0.336	0.310	0.265	
	taud	2.116	2.134	2.172	2.271	2.389	2.436	2.396	2.399	2.472	2.396	2.206	2.087	
	Ebn at Noon	497	680	789	825	859	866	834	806	775	663	454	365	
	Edh at Noon	46	78	102	111	106	103	105	98	78	62	44	33	
All-Sky Solar Radiation	RadAvg	0.25	0.83	2.27	3.84	5.36	5.55	5.45	4.19	2.55	1.04	0.28	0.14	
	RadStd	0.04	0.16	0.33	0.44	0.44	0.56	0.46	0.49	0.43	0.31	0.14	0.05	0.02

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	N/A	-158	-221	N/A	N/A
(47) Regional (47 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-152	N/A	N/A	N/A

Nomenclature: See separate page